

Force-directed placement

Force-directed place pulls each free cell toward the average of its net neighbors,
plus a weak center pull

The idea

- Each iteration
- Fixed pads stay put
- Too much alpha collapses the design; too little barely moves
- Report HPWL before and after with the same net model

Pseudocode

- Force-directed place blends each free cell toward its neighbor average with step α
- Fixed pads do not move
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- From starter fifty-two

Algorithm sketch — try these

INPUT: positions, nets, α , iters, fixed pads

OUTPUT: updated positions + HPWL

each iter, for free cell c :

$\text{tgt} \leftarrow \text{avg neighbor coords (+ weak center)}$

$\text{pos}[c] \leftarrow (1-\alpha) \cdot \text{pos}[c] + \alpha \cdot \text{tgt}$

pads stay fixed

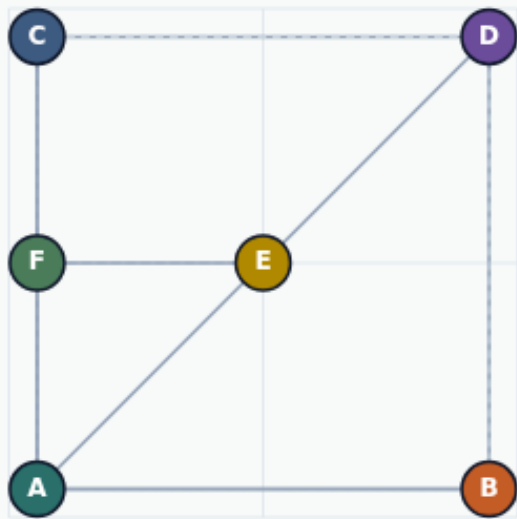
GOLDEN starter 52 $\rightarrow \approx 18.7$ after defaults

Start from HPWL fifty-two

FORCE-DIRECTED PLACE

Start from HPWL fifty-two

Step 1 / 5 · starter · module02-01-force-directed-place



Explanation

Force-directed place pulls free cells toward the average of their net neighbors, plus a weak center pull. Begin on the spread starter at HPWL fifty-two.

- Spring pull along net edges
- Weak center attraction
- Small alpha avoids collapse

HPWL: 52
iters: 5 (default)
alpha $\approx$ 0.12

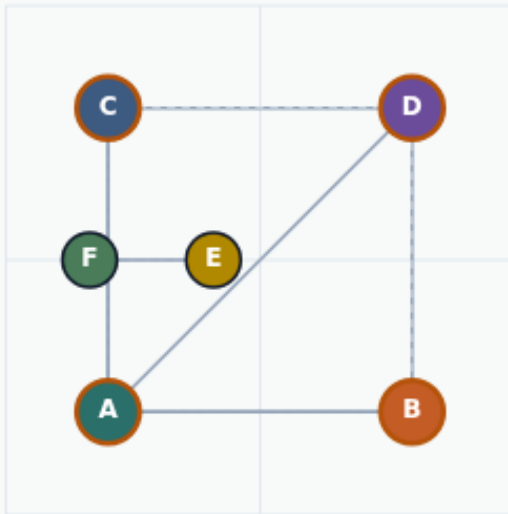
Start from HPWL fifty-two

Neighbors pull cells inward

FORCE-DIRECTED PLACE

Neighbors pull cells inward

Step 2 / 5 · mid-its · module02-01-force-directed-place



Explanation

After a couple of lite iterations, A-D drift toward the center while E and F follow their neighbors. Wirelength is already dropping.

- Each free cell → neighbor average
- Blend with small alpha
- Pads stay free in this lab

Mid HPWL: 34.5
Still above golden 14

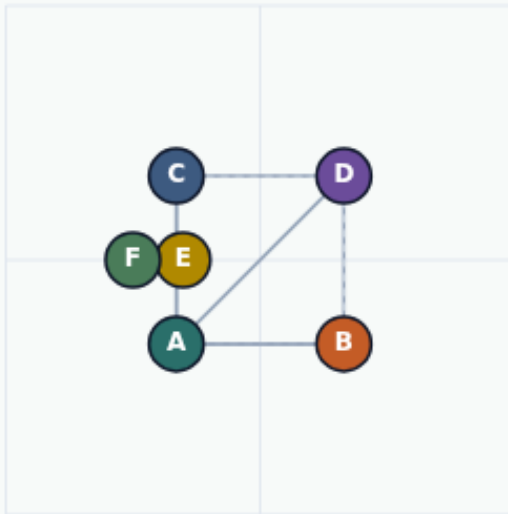
Neighbors pull cells inward

After force: about eighteen point seven

FORCE-DIRECTED PLACE

After force: about eighteen point seven

Step 3 / 5 · after-force · module02-01-force-directed-place



Explanation

Default five iterations land near eighteen point seven—clearly better than fifty-two, still above the compact golden fourteen.

- Report before/after HPWL
- Force ≈ 18.7 on this seed
- Not a full legalizer

Before: 52
After: 18.7
Measured: 18.7

After force: about eighteen point seven

Force vs golden compact

FORCE-DIRECTED PLACE

Force vs golden compact

Step 4 / 5 · vs-golden · module02-01-force-directed-place



Explanation

Golden fourteen is tighter still. Force is a cheap continuous move—good teaching progress without claiming the absolute minimum.

- Force ≈ 18.7
- Golden = 14
- Same nets, different tightness

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forceHpwlAfter: 18.7  
goldenHpwl: 14
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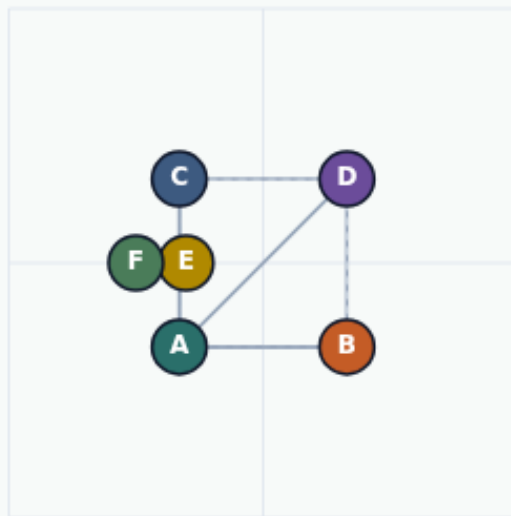
Force vs golden compact

Alpha trades speed vs collapse

FORCE-DIRECTED PLACE

Alpha trades speed vs collapse

Step 5 / 5 · takeaway · module02-01-force-directed-place



Explanation

Too much alpha stacks cells; too little barely moves. Lock the iteration count and alpha so your eighteen point seven golden stays stable.

- Neighbor average + center pull
- Starter 52 $\rightarrow$ ≈ 18.7
- Next: quadratic with fixed pads

forceHpwlAfter: 18.7

Alpha trades speed vs collapse

Browser lab track

- In the browser lab track, open the **force-directed-place** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse ``tiny_place.json``, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module